A Hybrid 2D VNet and ResNet Model for Multi-Organ Ultrasound Image Classification and Segmentation

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Abstract

Ultrasound imaging, with its non-invasive and real-time capabilities, plays a vital role in biomedical diagnostics. However, challenges in accurate classification and segmentation persist due to low resolution, noise, and complex tissue boundaries. This paper presents a hybrid model combining 2D VNet and 2D ResNet for multi-organ ultrasound image classification and segmentation. By leveraging VNet's segmentation strength and ResNet's feature extraction ability, the model shows high generalizability across various ultrasound tasks.The dataset includes ultrasound images from the breast, thyroid, liver, and kidney, with preprocessing steps such as Z-score normalization and image resizing. Data augmentation techniques were applied to enhance robustness. The model uses a combination of Dice and Cross-Entropy loss functions with the AdamW optimizer.Experimental results demonstrate the model's strong performance across different tasks and datasets, maintaining robustness in noisy data. The model's potential for clinical use in early tumor detection and improved diagnostic support for liver and kidney diseases is discussed.In conclusion, the hybrid 2D VNet and 2D ResNet model offers an effective solution for multi-task ultrasound image analysis, with promising applications in clinical settings.

1. Introduction

Ultrasound imaging, as a non-invasive, real-time, and cost-effective imaging technology, has been widely applied in the biomedical field. It provides real-time images of organs and tissues within the body, playing a crucial role in clinical diagnosis and tumor detection. However, due to the low resolution, high noise, and complex tissue boundaries in ultrasound images, tasks like classification and segmentation remain challenging. Most existing ultrasound image processing methods are tailored to specific organs or tasks and lack generalizability across various anatomical structures and pathological conditions.

To overcome these challenges, this paper proposes a model based on 2D VNet and 2D ResNet architectures to address multi-organ ultrasound image classification and segmentation tasks. The model incorporates the latest advances in deep learning and provides efficient image analysis results across multiple organs and disease tasks. Specifically, we designed a highly generalizable model that performs excellently across different ultrasound image tasks and can handle multi-organ classification and segmentation tasks simultaneously.

2. Methodology

2.1 Model Architecture

In this study, we adopted a hybrid model architecture combining 2D VNet and 2D ResNet. VNet is a convolutional neural network (CNN)-based segmentation network with a fully convolutional structure, which effectively captures spatial information and performs pixel-level segmentation. ResNet is a model that solves the challenges of training deep networks by introducing residual connections, which can effectively extract high-level features. By combining these two architectures, we aim to leverage the segmentation capabilities of VNet and the feature extraction abilities of ResNet.

2.1.1 2D VNet

2D VNet employs a U-Net style symmetric architecture, particularly suitable for medical image segmentation tasks. After each convolutional block, we use max-pooling operations to reduce the spatial resolution gradually. Then, up-sampling and convolutional layers are applied to restore the resolution to its original size for precise pixel-level predictions.

2.1.2 2D ResNet

2D ResNet uses residual learning, which reduces the vanishing gradient problem by introducing skip connections, making it possible to train deep networks effectively. ResNet is particularly well-suited for extracting complex image features, which is beneficial when dealing with the intricate tissue structures found in ultrasound images.

2.2 Datasets and Preprocessing

This study uses a dataset containing ultrasound images from different organs, including the breast, thyroid, liver, kidney, etc. All input images were Z-score normalized and resized to a fixed size of 512x512 to ensure consistency and comparability of the input data.

In terms of data augmentation, we applied random rotation, flipping, scaling, and Gaussian noise to enhance the model's robustness and generalizability. These data augmentation techniques effectively increase the diversity of the training data, helping to avoid overfitting.

2.3 Loss Function and Optimization Strategy

During training, we used a combination of Dice loss and Cross-Entropy loss to optimize the model. Dice loss effectively handles class imbalance issues in medical images, while Cross-Entropy loss helps optimize classification tasks. For optimization, we chose the AdamW optimizer, which combines the advantages of the Adam optimizer and weight decay strategies, helping to reduce overfitting and improve model performance.

2.4 Hyperparameter Tuning

To obtain the best model performance, we used random search for hyperparameter tuning, including the learning rate and weight decay. The initial learning rate was set to 1e-4, and we applied a cosine annealing strategy to dynamically adjust the learning rate. The training duration was 300 epochs with a batch size of 256.

2.5 Post-Processing Steps

To remove small connected components from the segmentation results, we applied a small connected component removal strategy. This helps clean up small regions caused by noise or incorrect segmentations, thereby improving the accuracy of the segmentation results.

3. Experimental Results

3.1 Datasets and Experimental Setup

We used 80% of the data for training, 10% for validation, and the remaining 10% for testing. The experiments were conducted on an A800 GPU, and the training took approximately 48 hours.

3.2 Performance Evaluation Metrics

To evaluate the performance of the model, we used metrics such as Dice coefficient, IoU (Intersection over Union), and accuracy. These metrics allowed us to comprehensively assess the model's performance across different tasks and datasets.

3.3 Result Analysis

The experimental results across multiple datasets showed that the hybrid model combining 2D VNet and 2D ResNet achieved excellent performance across all tasks, especially in complex ultrasound image segmentation tasks. Through data augmentation and post-processing strategies, the model maintained high robustness even on noisy real-world data.

4. Discussion

4.1 Model Advantages and Limitations

The model combines the strengths of VNet and ResNet, providing strong generalization capabilities for ultrasound image segmentation and classification tasks. However, despite achieving good results across multiple tasks, the model still faces some limitations when confronted with extreme noise or anomalous data. Future improvements could involve optimizing the network architecture further or incorporating more sophisticated attention mechanisms to enhance performance.

4.2 Potential Impact on Clinical Practice

If applied in clinical settings, this model could help clinicians improve diagnostic efficiency, reduce human errors, and provide more accurate treatment plans. Especially in early tumor detection and the auxiliary diagnosis of liver, kidney, and other diseases, the application of AI models would significantly improve clinical diagnostic outcomes.

5. Conclusion

This paper proposes a hybrid model based on 2D VNet and 2D ResNet architectures for ultrasound image analysis. The model is capable of handling multi-task processing across multiple organs and diseases. Experimental validation shows that the model performs well on different datasets and tasks, demonstrating strong generalization capabilities and robustness. Future work can explore additional data augmentation methods and optimization strategies to further enhance the model’s performance and reliability in real-world applications.

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